

AJAR — HCDS methodology pipeline

Panel 1 *Datasets*

GSM8K
n=50 deep-table • n=500 wide

StrategyQA
n=50 deep-table

Panel 2 *6 conditions per question (2 models × 3 prompts)*

Qwen3-4B Instruct

Qwen3-4B Thinking

Prompts
explicit_cot
explicit_no_cot
neutral_strict

Deterministic decoding (greedy) • caps: 1024 baseline / 1536 mech / 384 probe

Panel 3 *Per-question feature vector (6 dims, z-scored per model)*

- | | |
|-----------------------------|---------------------------------|
| 1. Latency per output token | ← timing |
| 2. Token entropy mean | ← attention probe over response |
| 3. Token entropy slope | ← attention probe over response |
| 4. Paraphrase consistency | ← build_paraphrases.py |
| 5. Perturbation Δ accuracy | ← build_perturbations.py |
| 6. Mechanistic Δ accuracy | ← anchor zero-out (n=50 only) |

Panel 4 *HCDS computation (per question)*

$f_{\text{neutral}}, f_{\text{no_cot}}, f_{\text{cot}}$ (one z-vector per prompt)
$$\text{HCDS}_q = D(f_{\text{neutral}}, f_{\text{no_cot}}) - D(f_{\text{neutral}}, f_{\text{cot}})$$

Euclidean distance • partial-feature aware

Panel 5 *Statistical testing (per model × dataset)*

Bootstrap CI
1000 resamples • seed=17
2.5% / 97.5% percentile

One-sample t-test
 $H_0: \text{HCDS} = 0$
two-sided

Panel 6 *Headline verdict*

GSM8K n=500	Instruct +2.38 (p=2e-141)	Thinking +0.33 (p=1.9e-7)
StrategyQA n=50	Instruct +1.87 (p=4e-12)	Thinking +0.60 (p=2e-3)